MULTILAYER DESIGN GUIDELINES 3.1

NCAB GROUP

Getting it right from the start

Nothing affects the PCB's cost and quality as much as the initial design. As modern electronic products are expected to offer more and more advanced functions, while the products themselves are becoming smaller and smaller, this puts greater demands on the PCB design and the importance of making the right choices at the design stage.

More than 30% of the Gerber data packs we receive do have some issues, generally, ambiguous information, errors, design rule conflicts, missing information and contradictions between the data and specifications. To prevent that it gets wrong from the start, we have put together these design guidelines, to use as a checklist.

When using combinations of parameters, you should always consult your local NCAB technical contact person.

Related NCAB Guidelines and Tools:
HDI Design Guideline
Stackups and Impedances Guideline
PCB Design Tips
PCB Design Checklist
PCB Cost drivers

* Talk to your local NCAB FAE if you need even better capabilities than shown under advanced. Combinations of parameters shown under advanced, or even exceeding advanced level, will require a thorough design review.

Design tips

1. If specifying the build, aim for a symmetrical stack-up in terms of dielectric spacing to reduce the risk of excessive bow & twist on the final board.
2. Internal copper plane layers shall be placed symmetrical in the stack-up, to minimize bow & twist
3. Avoid using cores as outer layer as this increase cost.
4. Use the same copper thickness on both sides of an inner layer core for smoother production.
5. Aim for similar amount copper area on top and bottom side to avoid over / under plating by using copper filling in empty areas.
6. Avoid tight hole tolerances when not needed.
7. Allow the factory to remove non-functional pads for via holes if uncritical. Read IPC-2222B, 9.1.4 table 9-3 for better understanding.
8. Use tear-drops in all layers, on all designs both for IPC class 2 and 3. This will help reliability and reduce the need of Engineering Questions.
9. Use balanced spacing between important components to avoid uneven copper foil and inconsistent PCB heat dissipation, resulting in bow & twist.
10. Utilize back drilling structures instead of sequential lamination to avoid unwanted via stubs. This will reduce pressing cycles / process time and also save cost.
11. Utilize thin trace widths only locally such as routing inside BGA pattern and other dense footprints.
12. Always prioritize trace separation before trace width, to make room for etch compensation in the factory.

Peelable & Carbon

	GENERAL	MODERATE	ADVANCED*
PEELABLE MASK (mm)			
Nominal Thickness	0.3-1.0	0.3-1.0	0.3-1.0
Max covered hole diameter	1.50	4.50	5.00
Min distance to not covered feature	1.00	0.50	0.50
KAPTON TAPE (mm)			
Nominal Thickness	0.05	0.05	0.05
CARBON INK (mm)			
Min space between carbon	0.60	0.35	0.30
Min carbon opening bigger than copper pattern per side	0.30	0.15	0.13
Min space between carbon edge to around copper	0.56	0.35	0.20
Position accuracy	0.20	0.10	0.08
Nominal thickness		'8-25µm	

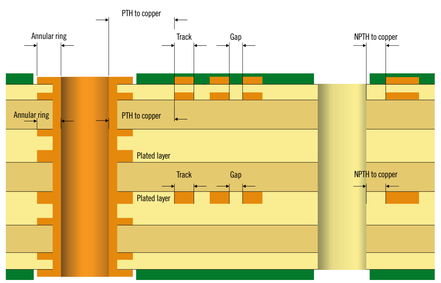
Surface finish

		Thickness of deposit	Solderability performance	Handling concerns	Fine pitch / Small BGA & SMD	Excellent flatness	Press fit	Suitable for peelable	Shelf / life concerns	Assembly rework	Wire bonding	Suitable for baking	Comments
Pb-F HASL	1-40	YYY				Y	Y		Y			Y	Bridging on fine pitch, avoid on thin pcb
SnPb HASL	1-40	YYY				Y	Y		Y			Y	Bridging on fine pitch, avoid on thin pcb
Flash gold	**Au:** 0.025-0.125 **Ni:** 3-6	Y		Y	Y		Y			Y (Alu)	Y		
OSP	0.2-0.65	Y	Y	Y	Y			Y					Short operating window, storage limits
ENIG	**Au:** 0.05-0.125 **Ni:** 3-6	Y	Y	Y	Y		Y		Y	Y (Alu)		Y	Avoid soldermask Defined bga
ENEPIG	**Ni:** 3-8µm **Pd:** 0.05-0.3µm **Au:** 0.05-0.125µm	Y	Y	Y	Y		Y		Y	Y (Alu)	Y		
Immersion Tin	1.0-1.2	Y	Y	Y	Y	Y							Aggressive to s-mask, use > 0,125mm dam
Immersion Silver	0.12-0.4	Y	Y	Y	Y	Y		Y					Short operating window, high speed pcb
OSP + Immersion Gold	**OSP:** 0.2-0.65 **Ni:** 3.0-6.0 **Au:** 0.05-0.125	Y	Y	Y	Y			Y					Short operating window, storage limits
Carbon	8-25	NA		NA		NA	Y		NA		Y		Contacts
Edge hard gold contacts	**Au:** 0.8-1.5 **Ni:** ≥ 2.5	NA		NA	Y	NA	Y		NA		Y		Contacts
Internal hard gold contacts	**Au:** 0.25-0.75 **Ni:** ≥ 2.5	NA		NA	Y	NA	Y		NA		Y		Contacts, need space to track out to edge

Circuitry

TRACK&GAP (mm)	GENERAL IL	GENERAL OL*	MODERATE IL	MODERATE OL*	ADVANCED* IL	ADVANCED* OL*
1/3 oz					0.075 / 0.075	0.075 / 0.075
1/2 oz	0.150 / 0.200	0.150 / 0.200	0.100 / 0.150	0.125 / 0.150	0.075 / 0.075	0.075 / 0.100
1 oz	0.150 / 0.200	0.200 / 0.250	0.125 / 0.150	0.150 / 0.175	0.100 / 0.100	0.125 / 0.150
2 oz	0.200 / 0.250	0.250 / 0.300	0.175 / 0.225	0.200 / 0.250	0.150 / 0.175	0.175 / 0.225
3 oz	0.250 / 0.300	0.300 / 0.350	0.225 / 0.275	0.250 / 0.300	0.200 / 0.250	0.225 / 0.275
ANNULAR RING OF VIAS (ACCEPT 90°BROKEN PER IPC CLASS 2) (mm)						
1/3 oz					0.100	0.100
1/2 oz	0.200	0.200	0.150	0.150	0.100	0.125
1 oz	0.200	0.200	0.150	0.175	0.125	0.150
2 oz	0.250	0.300	0.225	0.250	0.175	0.200
3 oz	0.300	0.400	0.275	0.350	0.225	0.250
PTH TO COPPER (mm)						
1/3 oz					0.175	0.200
1/2 oz	0.350	0.400	0.250	0.300	0.200	0.225
1 oz	0.350	0.450	0.250	0.350	0.225	0.300
2 oz	0.450	0.600	0.300	0.500	0.250	0.425
3 oz	0.550	0.750	0.400	0.650	0.300	0.525
NPTH TO COPPER (mm)						
1/3 oz					0.200	0.200
1/2 oz	0.300	0.300	0.250	0.250	0.200	0.200
1 oz	0.300	0.300	0.250	0.250	0.200	0.225
2 oz	0.400	0.400	0.300	0.300	0.225	0.250
3 oz	0.400	0.400	0.300	0.300	0.250	0.275

* Outer layers and plated Inner layers.



Wrap copper

IPC FEATURE	DEMAND (µm)	BASE COPPER (oz)	GENERAL TRACE	GENERAL GAP	MODERATE TRACE	MODERATE GAP	ADVANCED* TRACE	ADVANCED* GAP
Buried via ≥ 2L, (PTH and blind via)	**Wrap copper 5** (IPC Class 2 and 3)	**1/3**	N/A		0.12	0.12	0.10	0.10
		1/2	N/A		0.14	0.14	0.12	0.12
		1	N/A		0.15	0.15	0.14	0.14
Microvia (blind and buried)	**Wrap copper 5** (IPC Class 2 and 3)	**1/3**	N/A		0.12	0.12	0.10	0.10
		1/2	N/A		0.14	0.14	0.12	0.12
		1	N/A		0.15	0.15	0.14	0.14
Buried via cores (2L)	**Wrap copper 5** (IPC Class 2 and 3)	**1/3**	N/A		0.12	0.12	0.10	0.10
		1/2	N/A		0.14	0.14	0.12	0.12
		1	N/A		0.15	0.15	0.14	0.14

For HDI+VIA IN PAD TYPE VII+Blind Vias etc high technology, please contact NCAB for confirmation of features.

Base materials

	STANDARD	ADVANCED*
MATERIAL TYPE & THICKNESS		
Rigid	FR-4.0 Low, Mid & High Tg FR-4.1 (Halogen-free) Low, Mid & High Tg, CTI 100 – 249V **Standard – Loss material (Dk > 4,0 Df > 0,015)** SYL S1000-H, SYL S1000-2M, ITEQ IT-158, IT-180A, TUC TU-768 Panasonic R-1755V, Panasonic R-1566W, R-1566WN, Ventec VT-47, Isola 370HR, EMC EM-285, EMC EM-827, etc.. **Mid – Loss material (Dk 3.5–4.0 Df 0.010–0.015)** TUC TU862HF, EMC EM370D, ITEQ IT170GRA(1), Megtron-2, etc..	FR-4.0 Low, Mid & High Tg FR-4.1 (Halogen-free) Low, Mid & High Tg, CTI 250 – 600V, Automotive material FR-15.0 & FR15.1 **Low – Loss material (Dk 3.4–3.8 Df 0.005–0.010)** AGC N4000-13(series), Isola FR408HR, Megtron-4, Megtron-4S, SYL S7038, SYL S7439, TUC TU-863+, TUC TU-872SLK (series), EM-526,EMC EM-888, Isola I-Speed, etc.. **Very Low – Loss material (Dk 3.2–3.6 Df 0.003–0.005)** AGC MW1000 & MW2000 & MW3000, Megtron 6 & 6N, EMC EM-891 & EM-528, SYL Synamic6, Isola I-Tera MT40, Rogers RO4350B & RO3000(series), RF-35, RF-35A2, TLX (se-ries), AD250, FL-700LD, ITEQ IT-968(series), TUC TU-883, etc.. **Ultra Low – Loss material (Dk 3.0–3.4 Df 0.002–0.003)** AGC MW4000, Panasonic R-5785, TUC TU- 933E, EMC EM-890K, ITEQ IT-988GSE, etc.. **Extremely Low – Loss material (Df. <0.002)** Panasonic R-5795 (N), R-5795 (U), EMC EM-892K
IMS	Bergquist MP, HT & CML ITEQ T-Lam, Laird TLAM SS Taiflex, Dupont FR & AP, Panasonic, ShengYi, Doosan.	IMS Al & Cu based Including: Bergquist HPL, Ventec VT, Polytronics TCB, Doosan DST, Denka, Arlon, Chin-Shi
THICKNESS (mm)		
Min. dielectric	0.05mm for PCB 0.025mm for FPC	0.025mm for PCB 0.012mm for FPC
COMMON PREPREG THICKNESS (TOLERANCE AS PER IPC-4101 B/L) (mm)		
7628 (50%)	0.22	0.22
7628 (43%)	0.18	0.18
2116 (52%)	0.12	0.12
2116 (56%)	0.13	0.13
1080 (65%)	0.08	0.08
106 (75%)	0.05	0.05

Standard core thickness include 0.10, 0.20, 0,4, 0.6, 0.8 and 1.00mm.
Please contact NCAB Group for full details reg.rding material availability.

Layer count

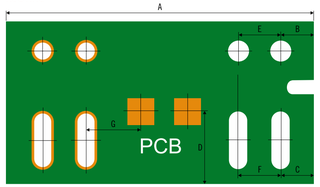
	GENERAL	MODERATE	ADVANCED*
LAYER COUNT (Layers)			
Rigid ML	4-8	10-18	20+

Board thickness

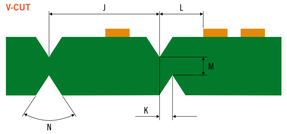
	GENERAL	MODERATE	ADVANCED*
BOARD THICKNESS (mm)			
4L	0.6-2.5	0.5-3.2	0.2-8.0
6-10L	0.7-2.5	0.6-6.0	0.5-8.0
12L +		1.3-6.0	1.2-8.0

Profile

ROUTING



V-CUT

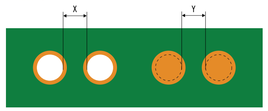


	GENERAL	MODERATE	ADVANCED*
ROUTING (mm)			
A: Routing	+/-0.13	+/-0.10	+/-0.10
B: Hole to edge	+/-0.15	+/-0.13	+/-0.10
C: Slot to edge	+/-0.15	+/-0.13	+/-0.10
D: Pattern to edge	+/-0.15	+/-0.13	+/-0.10
E: Hole to hole	+/-0.13	+/-0.10	+/-0.075
F: Slot to Slot	+/-0.13	+/-0.10	+/-0.075
G: Slot to Pattern	+/-0.13	+/-0.10	+/-0.075
Recommened pattern to edge	0.50	0.30	0.20
Maximum PCB size	457 x 508	580 x 680	580 x 1080
V-CUT (mm)			
J: Position accuracy (parameters based on 30° V-cut angle and PCB thickness 1,6 mm)	+/-0.15	+/-0.13	+/-0.10
K: Blade offset	+/-0.15	+/-0.13	+/-0.10
L: Keep out area (dimension)	0.5	0.45	0,4
M: Web thickness (@ 1,60mm)	0.45 +/-0.10	0.45 +/-0.10	0,45 +/-0.10
N: Angle	30°	20°, 30°	20°, 30°, 45°, 60°
Min board thickness	0.8	0,4	0,3
Max board thickness	2.3	3,2	8,0

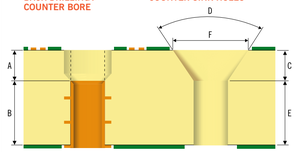
For boards < 0.50mm, or > 2.30mm thick it is not recommended to use scoring / V-Cut.

Drilling & hole sizes

DRILLING



BACK DRILLING / COUNTER BORE — COUNTER SINK HOLES

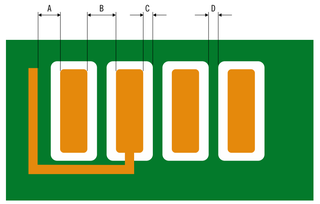


	GENERAL	MODERATE	ADVANCED*
DRILLING (mm)			
PTH tolerance (< 6.35 mm Ø) *	(±) 100µm	(±) 75µm	(±) 50µm
NPTH tolerance	(±) 75µm	(±) 50µm	(±) 25µm
PTH slot's width	(±) 127µm	(±) 100µm	(±) 75µm
PTH slot's length	(±) 127µm	(±) 100µm	(±) 100µm
NPTH slot's width (drilled)	(±) 100µm	(±) 75µm	(±) 50µm
NPTH slot's length (drilled up to 8 mm length)	(±) 100µm	(±) 75µm	(±) 75µm
X: Min. space between via hole walls in same net	0,3	0,25	0,15
X: Min. space between via hole walls in different net	0,4	0,35	0,25
Y: Min. space between IPC-4761 type VII via hole walls in same net	0,5	0,45	0,35
Y: Advanced capability 0.45mm hole edge to hole edge is a little big, we can change it to 0.4mm	0,6	0,55	0,45
BACK DRILLING / COUNTER BORE (mm)			
A: Depth tolerance	+/-0.20	+/-0.15	+/-0.10
B: Min. remain thickness and tolerance	0.5+/-0.25	0.35+/-0.1	0.3+/-0.1
COUNTER SINK HOLES (mm)			
C: Depth tolerance	+/-0.20	+/-0.15	+/-0.1
D: Angle and tolerance	90+/-10°	82, 90+/-10°	82, 90, 135+/-10°
E: Min. remain thickness and tolerance	0.50 +/-0.25	0.35 +/-0.1	0.3 +/-0.1
F: Countersink tolerance	+/-0.40	+/-0.15	+/-0.10

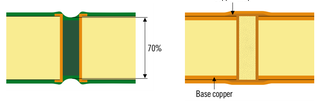
Holes > 6.35 mm Ø are routed and may have wider tolerances depending on hole size.

Soldermask / Via hole plugging

SOLDERMASK



VIA HOLE PLUGGING



Type VI soldermask filled via hole. Target 100% fill. Acceptable ≥ 70% fill of via.

Type VII resin filled via hole (over plated / capped).

	GENERAL	MODERATE	ADVANCED*
COLORS (mm)	**Gloss:** green / black / white / red / blue **Matte:** green / black	**Gloss:** green / black / white / red / blue / yellow **Matte:** green / black / red / blue	**Gloss:** green / black / white / red / blue / yellow / transparent **Matte:** green / black / red / blue / yellow / white **Semi matte:** green
SOLDERMASK THICKNESS (mm) 10-30µm on copper surface, 5µm on copper edge			
A: SMT to covered copper	0.20	0.15	0.10
B: Copper to copper spacing	0.24	0.20	0.18
C: Solder mask oversize	0.08	0.06	0.05
D: Minimum soldermask web	0.10	0.08	0.075

For >1oz base copper, please contact NCAB for confirmation of features.

	GENERAL	MODERATE	ADVANCED*
VIA HOLE PLUGGING VS HOLE SIZES (mm)			
Soldermask IPC 4761 Type VI	0.30-0.50	0.25-0.50	0.20-0.60
Resin non conductive IPC4761 Type VI		0.25-0.50	0.20-0.50
Over plated/capped IPC 4761 type VII		0.20-0.50	0.20-0.50
Resin electrical conductive			0.20-0.50
Resin thermal conductive			0.20-0.50

When more than one hole diameter shall be plugged, diameter variation shall be less than 0,2 mm.

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